The documentation and process conversion measures necessary to comply with this revision shall be completed by 14 September 2012.

INCH-POUND

MIL-PRF-19500/678A
14 June 2012
SUPERSEDING
MIL-PRF-19500/678
1 July 2000

* PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON,DUAL SCHOTTKY CENTER TAP POWER RECTIFIER, SURFACE MOUNTED, TYPES 1N6840U3 AND 1N6841U3, JAN, JANTX, JANTXV, JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, dual Schottky power rectifier in a surface mount package. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (U3, similar to SMD.5).

* 1.3 Maximum ratings. Unless otherwise specified, T_C = +25°C.

Type	V_{RWM}	I_O (1) T_C= +100 °C	I_{FSM} T_C =+25 °C t_p= 8.3 ms	C_J at 5 V	$R_{\theta JC}$ (2)	T_{STG} and T_J
	V	A dc	A (pk)	pF	°C/W	°C
1N6840U3	35	10	200	400	2.8	-65 to +150
1N6841U3	45	10	200	400	2.8	-65 to +150

(1) Derate linearly at 200 mA/°C from $T_J = T_C$ =+100°C to + 150°C.
(2) Each individual diode.

* 1.4 Primary electrical characteristics. $R_{\theta JC}$ = 1.7°C/W both legs tied together.

* Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil/.

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2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

* DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

* DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or https://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

* 2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

* 3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

* 3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 herein.

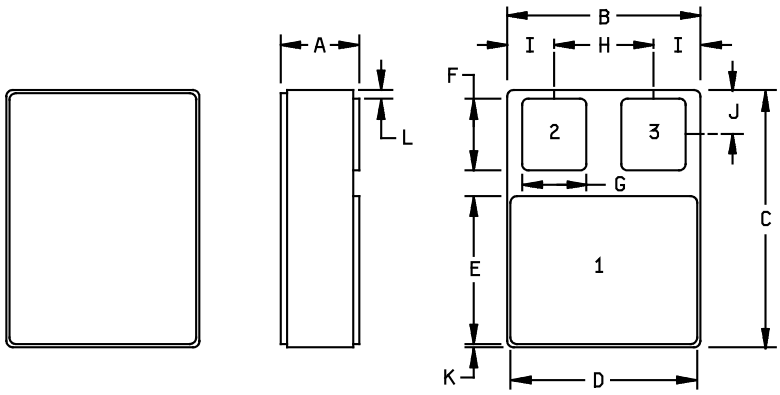
3.4.1 Polarity. Polarity and terminal configuration shall be in accordance with figure 1 herein.

* 3.4.2 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

* 3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4 and table I herein.

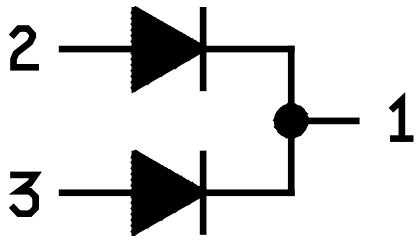
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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	0.111	0.122	2.82	3.10
B	0.291	0.301	7.39	7.65
C	0.395	0.405	10.03	10.29
D	0.281	0.291	7.14	7.39
E	0.220	0.230	5.59	5.84
F	0.115	0.125	2.92	3.18
G	0.090	0.100	2.29	2.54
* H	0.125	0.135	3.18	3.43
I	0.073 TYP.	-	1.85 TYP.	-
J	0.083 TYP.	-	2.11 TYP.	-
K	0.005 TYP.	-	0.13 TYP.	
L	0.015 TYP.	-	0.38 TYP.	-

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Terminal 1 is common cathode.
4. Terminal 2 is anode 1.
5. Terminal 3 is anode 2.

SCHEMATIC



* FIGURE 1. Physical dimensions and configuration (U3, similar to SMD.5).

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* 3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

* 4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

* c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

* 4.3 Screening (JANTX, JANTXV and JANS levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table E-IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
(1) 3c	Method 3101 of MIL-STD-750, (see 4.3.2).	Method 3101 of MIL-STD-750, (see 4.3.2).
9 and 10	Not applicable.	Not applicable.
11	V_{F2} and I_{R1}.	V_{F2} and I_{R1}.
12	Condition A. See 4.3.1, t = 240 hours.	Condition A. See 4.3.1, t = 48 hours.
13	Subgroups 2 and 3 of table I herein; ΔV_{F2} = ±50 mV, ΔI_{R1} = 100 percent of initial value or 25 µA, whichever is greater.	Subgroup 2 of table I herein; ΔV_{F2} = ±50 mV, ΔI_{R1} = 100 percent of initial value or 25 µA, whichever is greater.

* (1) Shall be performed anytime after temperature cycling, screen 3a. JANTX and JANTXV levels do not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. Burn-in conditions are as follows: Method 1038 of MIL-STD-750, test condition A. T_C = + 125°C; V_R = 80 percent of rated V_{RWM} dc.

* 4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_M, t_{SW}, (and V_H where appropriate). Measurement delay time (t_{MD}) = 100 µs max. See table III, group E, subgroup 4 herein.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

* 4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table E-V of MIL-PRF-19500, and table I herein. Electrical measurements (end points) shall be as specified in table I subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JAN, JANTX and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end points) shall be in accordance with table I subgroup 2 herein. Delta requirements shall be in accordance with table II herein.

4.4.2.1 Group B inspection, table E-VIA (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	4066	I_{FSM} = 200 A; 1 surge of 8.3 ms superimposed on I_O. Condition A, T_A = 25°C; V_R = 0; I_O = 10 A continuous half-wave.
B4	1037	Each diode: I_F or I_O = 2 A (minimum) ΔT_J = 85°C, + 15°C, - 5°C for 2,000 cycles minimum.
B5	1038	Condition B, T_{JMAX} = +150°C, T_A = 100°C, 240 hrs.
B6	3101	Limit for thermal resistance is 2.80ºC/W per side.

* 4.4.2.2 Group B inspection, table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B2	4066	I_{FSM} = 200 A; 1 surge of 8.3 ms superimposed on I_O. Condition A, T_A = 25°C; V_R = 0; I_O = 10A continuous half-wave.
B3	1037	Each diode: I_F or I_O = 2 A (minimum) ΔT_J = 85°C, +15°C, - 5°C for 2,000 cycles minimum.
B5		Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows. Electrical measurements (end points) shall be in accordance with table I subgroup 2 herein. Delta requirements shall be in accordance with table II herein.

Subgroup	Method	Condition
C3	2006	Conditions are as follows: X_1, Y_1, Z_1 and Z_2 axis.
C5	3101	See 4.3.2, $R_{\theta JC}$ = 2.80°C/W maximum, per side.
C6	1037	Each diode: I_F or I_O = 2 A (minimum) ΔT_J = 85°C, +15°C, -5°C for 6,000 cycles minimum.

4.4.4 Group E inspection. Group E inspection shall be performed for qualification or requalification only. The tests specified in table III herein must be performed to maintain qualification.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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* TABLE I. Group A inspection.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance 3/	3101	See 4.3.2	$Z_{\theta JX}$			°C/W
Forward voltage	4011	I_{FM} = 3 A (dc) pulsed (see 4.5.1). I_{FM} =10 A (dc) pulsed (see 4.5.1). I_{FM} =20 A (dc) pulsed (see 4.5.1).	V_{F1} V_{F2} V_{F3}		0.62 0.75 0.88	V dc V dc V dc
Reverse current leakage 1N6840U3 1N6841U3	4016	DC method, (see 4.5.1.) V_R = 35 V dc V_R = 45 V dc	I_{R1}		100	µA dc
Subgroup 3						
High temperature operation:		T_A = +100°C				
Reverse current leakage 1N6840U3 1N6841U3	4016	DC method, pulsed (see 4.5.1). V_R = 35 V dc V_R = 45 V dc	I_{R2}		15	mA dc
Forward voltage	4011	I_F = 10 A pulsed (see 4.5.1)	V_{F4}		0.63	V dc
		I_F = 20 A pulsed (see 4.5.1)	V_{F5}		0.70	
Forward voltage	4011	T_A = -55°C Pulsed (see 4.5.1), I_F = 10 A (dc).	V_{F6}		0.85	V dc

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Junction capacitance	4001	V_R = 5 V dc; f = 1 MHz, V_{SIG} = 50 mV (p-p) (max)	C_J		400	pF
Subgroup 5						
Insulation resistance (Dielectric withstanding voltage)	1016	V_R = 600 V dc; from lid to bottom case. All terminals shorted.	D_{WV}		10	µA
Subgroup 6 and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.
2/ All measurements are for each side.
3/ If this test is performed 100 percent in screening, this test need not be performed in group A.

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* TABLE II. Groups B and C delta measurement. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward voltage	4011	I_F = 10 A (pk) pulsed (see 4.5.1)	ΔV_{F2}		±50	mV
2.	Reverse current leakage	4016	DC method $V_R = V_{RWM}$	ΔI_{R1}		25 µA or 100 percent of initial reading, whichever is greater.	

1/ The electrical measurements for table E-VIA (JANS) of MIL-PRF-19500 are as follows:
- a. Subgroup 3, see table II herein, steps 1 and 2.
- b. Subgroup 4, see table II herein, steps 1 and 2.
- c. Subgroup 5, see table II herein, steps 1 and 2.

2/ The electrical measurements for table E-VIB (JAN, JANTX and JANTXV) of MIL-PRF-19500 are as follows:
- a. Subgroup 2, see table II herein, steps 1 and 2.
- b. Subgroup 3, see table II herein, steps 1 and 2.
- c. Subgroup 6, see table II herein, steps 1 and 2.

3/ The electrical measurements for table E-VII of MIL-PRF-19500 are as follows:
- a. Subgroups 2 and 3, see table II herein, steps 1 and 2 for all levels.
- b. Subgroup 6, see table II herein, steps 1 and 2 for all levels.

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* TABLE III. Group E inspection (all quality levels) for qualification only.

	Inspection	MIL-STD-750		Qualification Inspection
		Method	Conditions	
	Subgroup 1			38 devices, c = 0
	Thermal shock (temperature cycling)	1051	500 cycles. Condition G.	
*	Hermetic seal Fine leak Gross leak	1071		
	Electrical measurements		See table I, subgroup 2.	
	Subgroup 2			38 devices, c = 0
	Burn-in (steady-state blocking life)	1038	t = 1,000 hours, T_J = + 125°C; V_R = 80 percent of rated V_{RWM}.	
	Electrical measurements		See table I, subgroup 2.	
	Subgroup 3			
	Not applicable			
	Subgroup 4			Sample size N/A
*	Thermal impedance curves	3101	See MIL-PRF-19500.	
	Subgroup 5			5 devices, c = 0
	Surge	4066	Condition A, T_A = +25°C I_{FSM} = 200 A, one surge of 8.3 ms superimposed on I_O. V_R = 0; I_O = 10 A pk half sine wave, continuous.	
	Electrical measurements		See table I, subgroup 2.	

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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

* 6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.daps.dla.mil .

* 6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Navy - EC
NASA - NA
DLA - CC

Preparing activity:
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